SHIELD — Systematic Hyperlocal Income-Event Loss Detection

> Parametric micro-insurance for Q-commerce delivery riders in India.
> Zero paperwork. Automated triggers. UPI payout in under 4 hours.

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1. Problem Statement

Q-commerce delivery riders (Blinkit, Zepto, Swiggy Instamart) operate on a 10–20 minute SLA promise. Their income is entirely event-driven — they earn per delivery, with zero fixed pay. A single disruption event — a flash flood, a bandh, a dark store power outage, a VIP convoy — can erase 100% of their earnings for an entire shift with no recourse.

Existing insurance products do not cover income loss. They cover death, hospitalisation, and vehicle damage — but not the most frequent pain point: the shift you couldn't work, through no fault of your own.

SHIELD is a weekly parametric micro-insurance product that pays out automatically, via UPI, within 4 hours of a verified disruption event — no claim forms, no surveys, no waiting.

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2. Persona-Based Scenarios & Workflow

Persona A — Rajan, Blinkit Rider, Bengaluru

Profile: 28 years old. Works 6-hour evening shifts (5 PM – 11 PM). Average weekly earnings: ₹4,200. Pays ₹49/week for SHIELD.

Scenario: Flash flood, Koramangala, 7:45 PM Friday

Rajan is en route to deliver a grocery order when the underpass on 80 Feet Road floods to 15cm depth within 12 minutes of a cloudburst. His app shows zero new assignments — the dark store

has throttled all outgoing orders.

Step	What Happens
7:51 PM	IMD rainfall API records 22mm/hr for Bengaluru South district. SHIELD's disruption classifier fires an auto-trigger.
7:52 PM	Google Maps traffic speed API for the zone drops to 4 km/hr. Windy.com 5-min radar confirms localised cell. Dark store assignment rate = 0. Three signals corroborate.
7:53 PM	SHIELD pushes Rajan a notification: *"Disruption detected in your zone. Your earnings are protected. Tap to confirm you're active."*
7:54 PM	Rajan taps confirm. His GPS geofence is verified within the affected zone. Claim opens automatically.
10:30 PM	Rain clears. Rajan resumes deliveries. Earns ₹310 in the final 90 minutes of shift.
11:05 PM	Claim quantum calculated: Baseline ₹700 (Friday 7–10 PM slot) × disruption fraction 0.9 × peak coefficient 0.95 = ₹598 expected loss. Actual earnings gap = ₹598 − ₹0 (during rain) = **₹598 payout queued.**
11:09 PM	₹598 credited to Rajan's UPI ID. Notification confirms amount and claim reference.

Persona B — Divya, Zepto Rider, Chennai

Profile: 24 years old. Works split shifts — 8 AM–12 PM and 6 PM–9 PM. Average weekly earnings: ₹3,600. Pays ₹42/week for SHIELD.

Scenario: State-level bandh, political party-called shutdown, Wednesday

Divya arrives at her dark store at 8 AM to find it locked — the store manager couldn't open due to party workers blocking the commercial complex entry. The platform app shows no assignments available across Chennai.

Step	What Happens
6:30 AM	SHIELD's news NLP parser detects three verified news outlets (ANI, The Hindu, Manorama) publishing the bandh declaration. Confidence score: 0.97. Auto-trigger fires for Chennai district.
8:05 AM	Divya opens the app. A full-shift protection card is displayed proactively — she doesn't need to file anything.
8:10 AM	Dark store assignment rate for her hub's pin code = 0. Peer consensus: 11 of 12 riders in her 500m radius are showing idle status. All signals confirmed.
End of day	Full shift payout calculated: Morning baseline ₹480 + Evening baseline ₹360 = **₹840 payout.**
7:00 AM Thursday	₹840 credited via the overnight batch sweep, before Divya starts her next shift.

Persona C — Suresh, Swiggy Instamart Rider, Hyderabad

Profile: 35 years old. Works 10 AM – 4 PM. Average weekly earnings: ₹2,900. Pays ₹35/week for SHIELD.

Scenario: VIP convoy road closure + partial delivery recovery

A Union Minister's visit seals a 2.5 km stretch of Jubilee Hills road for 70 minutes. Suresh is mid-shift and physically cannot reach his next 4 delivery addresses. He manages to complete 3 deliveries on an alternate route, earning ₹180 during the window.

Step	What Happens
12:18 PM	Traffic speed API for Jubilee Hills drops to 2 km/hr. No IMD or official signal available — this is a semi-auto trigger scenario.
12:19 PM	SHIELD pushes: *"Unusual traffic detected in your zone. Did a road closure affect your shift? Tap to start a claim."*
12:21 PM	Suresh taps confirm. He uploads a 25-second geo-tagged video of the police cordon. Enters the claim lane.
2:45 PM	AI reviewer (human-in-loop) verifies video, cross-references Google Maps historical traffic anomaly log for the pin code + 70-minute timestamp window. Claim approved.
Calculation	Baseline ₹580 (Tuesday 12–2 PM slot) × disruption fraction (70/120 min) × coefficient 0.65 = ₹222 expected loss. Actual earnings gap = ₹222 − ₹180 earned = **₹42 payout.**
3:10 PM	₹42 credited. Small number — but accurate. Suresh did work through part of the disruption, and the system tops up only his actual deficit.

> The tiered deficit model is intentional. Riders always benefit from working — compensation never exceeds what they would have earned, ensuring there is no incentive to stop working during a disruption.

Application Workflow (End-to-End)

```
Onboarding
  └── Rider downloads SHIELD app
  └── KYC: Aadhaar + Selfie verification (Digilocker API)
  └── Bank penny-drop verification (UPI handle)
  └── Links delivery platform account (OAuth or API token)
  └── Selects weekly premium tier → Policy activates Monday 00:00

During Shift
  └── SHIELD runs passively in background
  └── Monitors: GPS location, dark store assignment rate, IMD/AQI feeds
  └── Disruption detected → Classify → Auto / Semi-auto / Manual lane

Claim Processing
  └── Auto: No action needed from rider
  └── Semi-auto: One-tap confirmation
  └── Manual: 30-sec geo-tagged photo/video + description

Payout
  └── Fraud firewall clears → Job queue → UPI payout attempt
```

```
  └── Retry logic (3 attempts, exponential backoff) → NEFT fallback
  └── 3 AM batch sweep → 7 AM credit for all overnight-pending payouts

Rider Dashboard
  └── Weekly earnings baseline (rolling 4-week median)
  └── Active policy status
  └── Claim history + payout log
  └── Premium deduction record
```

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3. Weekly Premium Model & Parametric Triggers

Premium Model

Premiums are dynamic, calculated weekly, and deducted on Monday morning from the rider's SHIELD wallet (pre-loaded or auto-debit from linked account).

Base Formula:

```
Weekly Premium = Base_Rate × Zone_Risk_Score × Earnings_Band_Multiplier × Platform_Modifier
```

Variable	Description
`Base_Rate`	₹35–₹75 flat range depending on city tier
`Zone_Risk_Score`	0.7–1.4 multiplier based on historical disruption frequency for the rider's primary dark store pin code (flood history, bandh frequency, AQI events, traffic accident density)
`Earnings_Band_Multiplier`	0.8–1.2 based on the rider's rolling weekly earnings — higher earners pay slightly more, proportional to the income at risk
`Platform_Modifier`	Blinkit (1.0) / Zepto (1.05) / Instamart (0.95) — reflects historical claim rates per platform from beta data

Indicative premium ranges:

City	Rider Earnings Band	Weekly Premium	Max Weekly Payout
Bengaluru	₹3,500–₹5,000	₹42–₹58	₹4,500
Chennai	₹3,000–₹4,500	₹38–₹52	₹4,000
Delhi NCR	₹4,000–₹6,000	₹55–₹75	₹5,500
Hyderabad	₹2,800–₹4,200	₹35–₹49	₹3,800

The maximum weekly payout is capped at 90% of the rider's 4-week median weekly earnings — preventing full income replacement while covering genuine losses.

Parametric Triggers — Priority Tier Map

Triggers are classified into three tiers based on verifiability and automation confidence.

Tier 1 — Hard Auto-Triggers (No Rider Action Required)

Trigger	Data Source	Threshold	Scope
Heavy Rainfall	IMD District API	>15mm/hr	Zone
Flash Flood (road depth)	BBMP/GHMC flood sensor network	>10cm road depth	Hyperlocal
Cyclone/Depression Warning	IMD Alert API	Orange/Red alert	City
Extreme AQI	CPCB AQI API	>400 (Severe)	City/Zone
Declared Bandh	Verified news NLP (ANI, PTI, The Hindu)	Confidence >0.90	City/Zone
Platform App Outage	Blinkit/Zepto internal API (partner integration)	Outage >15 min during peak	City
Dark Store Power Outage	Dark store IoT sensor (partner integration)	Outage >20 min	Store
Curfew/Section 144	State Government Gazette API	Any declared instance	Zone

Tier 2 — Semi-Auto Triggers (One-Tap Rider Confirmation)

Trigger	Corroboration Required (2 of 3)
VIP Convoy Road Closure	Traffic speed API <5 km/hr + Peer idle consensus + Manual photo
Film Shoot Road Block	Police permission public feed + Traffic speed + Peer consensus
Religious Procession	Known calendar event + Traffic speed + Peer consensus
RWA Delivery Blackout	Peer consensus + Dark store assignment rate drop + Rider geofence
Dark Store Picker Shortage	Assignment rate drop + Rider present at store geofence + Peer idle

Tier 3 — Manual Claim Lane (Photo/Video + AI Review, 2-hr SLA)

Absurd but legitimate scenarios requiring human-in-loop AI review:

- Stray cattle/animal mob blocking route (geo-tagged video required)
- Wedding baraat occupying lane (photo + approximate duration)
- Unexploded ordnance exclusion zone (news corroboration + geofence)
- Sewage main collapse road closure (municipal emergency feed + photo)
- Wildfire smoke/crop burn AQI spike (CPCB AQI + geofence)
- False fire alarm building evacuation (duration log + building geofence)
- Gas pipeline rupture evacuation zone (municipal/fire department feed)

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4. Platform Choice: Mobile vs Web

Decision: Mobile-first (Android PWA → Native Android App)

Rationale

Q-commerce delivery riders in India are almost entirely mobile-native. Smartphone penetration among gig workers is >95%; laptop/desktop access is near zero. Every key workflow — shift tracking, GPS verification, claim submission, UPI payout — is mobile-native by design.

Consideration	Web App	Mobile (Android)	Decision
Background GPS tracking	Not reliable	Full support	✅ Mobile
Push notifications for auto-triggers	Limited (browser required open)	Native, always-on	✅ Mobile
UPI deep link for payment confirmation	Awkward in browser	Seamless intent-based	✅ Mobile
Offline functionality (poor network zones)	Service worker limited	Native offline cache	✅ Mobile
Geo-tagged photo/video for manual claims	Browser Camera API (unreliable)	Native camera with metadata	✅ Mobile
Distribution in market	No install friction	Play Store familiarity	Neutral

Phase 1 approach: React Native PWA deployed as an installable Android app (no Play Store review cycle needed in MVP). Graduates to full native Android in Phase 2 once core flows are validated.

A lightweight web dashboard is built in parallel — but exclusively for:

- Rider onboarding (first-time KYC on a larger screen)
- Ops/admin claim review panel
- Partner (dark store operations) integration portal

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5. AI/ML Integration

5.1 — Earnings Baseline Engine (Regression Model)

Problem: A fair baseline cannot simply be the rider's raw historical average — past disruptions contaminate the baseline.

Solution: A gradient-boosted regression model trained on:

- Rider's own earnings history (rolling 12 weeks, same day/slot pairs)
- Zone-level demand index from platform data
- Day-of-week, time-of-day, and seasonal patterns
- Festival calendar effects (Diwali +30%, post-bandh recovery surge, etc.)

The model normalises out disruption-contaminated weeks before computing the baseline. Output: a per-slot expected earnings figure with a confidence interval. The lower bound of the interval is used as the payout denominator — conservative and fraud-resistant.

5.2 — Disruption NLP Classifier (News + Social Feed Parsing)

Problem: Bandhs, VIP movements, and protest-related closures are announced inconsistently — sometimes on Twitter, sometimes in a regional news outlet, sometimes only in a WhatsApp group.

Solution: A fine-tuned NLP classifier (base: IndicBERT or mBERT for multilingual Indian language support) that:

- Ingests RSS feeds from ANI, PTI, The Hindu, Manorama, Eenadu, Lokmat, and 40+ regional outlets
- Monitors Twitter/X keyword streams for disruption signals by city + pin code
- Scrapes public Telegram channels of key resident welfare associations
- Classifies each signal: disruption type, geographic scope, confidence score, expected duration
- Only fires a trigger when confidence > 0.90 and at least one corroborating signal exists

5.3 — Fraud Detection Engine (Ensemble Model)

Six parallel checks feed a gradient-boosted fraud score (0–1 scale). Claims with score >0.7 go to human review; >0.9 are auto-rejected with rider notification.

Check	Signal	Weight
Peer Consensus	% of riders in 500m radius showing idle status	0.30
Geofence Integrity	Rider GPS within zone at claimed time	0.20
Historical Claim Pattern	90-day claim frequency vs. zone average	0.20
Earnings Velocity Anomaly	Abnormal earnings immediately before/after window	0.15
Dark Store Assignment Rate	Platform API: assignments issued during window	0.10
Zone Corroboration	Google Maps traffic speed + Windy.com + Social signals	0.05

The model is retrained weekly on new labelled claim data. Human reviewers label edge cases in the Manual lane — these become high-quality training examples for the semi-auto tier over time, continuously tightening the automation boundary.

5.4 — Premium Pricing Model (Actuarial ML)

A time-series survival model trained on:

- Historical claim frequency by zone, season, platform, and earnings band
- IMD historical weather data (monsoon onset dates, flood event frequency by district)
- City-level bandh frequency (sourced from news archive)
- Dark store operational failure rate (partner data)

Output: expected annual claim value per rider segment, from which weekly premiums are back-calculated with a target loss ratio of 65–70% (leaving 30–35% for platform operations, reinsurance, and IRDAI-mandated reserve).

Premiums are re-priced every 4 weeks per rider based on updated zone risk scores and the rider's individual claim history — a light credibility weighting (no more than ±15% adjustment from zone base rate) to prevent over-personalisation on sparse data.

5.5 — Manual Claim AI Review (Vision + LLM)

For Tier 3 manual claims, the submitted photo/video is processed by:

1. A computer vision model verifying that the image contains a genuine physical blockage (road obstruction, flood water, animal, police cordon) — not a screenshot or fabricated image
2. EXIF metadata validation confirming geo-tag and timestamp match the claimed zone and window
3. An LLM (Claude API) that reads the rider's description, the visual model output, and all corroborating data signals, then produces a structured claim assessment with a recommendation (approve / reject / escalate)

Human reviewers see the LLM's recommendation with full reasoning. Their override decisions are logged and feed back into model improvement.

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6. Tech Stack

Mobile App

Layer	Technology
Framework	React Native (Expo managed workflow → bare workflow in Phase 2)
State Management	Zustand
Maps & Geofencing	react-native-maps + background geolocation (transistorsoft)
Offline Support	WatermelonDB (local SQLite)
Push Notifications	Firebase Cloud Messaging (FCM)

Layer	Technology
UPI Deep Linking	react-native-upi-pay

Backend / API

Layer	Technology
API Layer	FastAPI (Python) — async, OpenAPI auto-docs
Auth	Supabase Auth (JWT) + Aadhaar OTP via Digilocker API
Database	PostgreSQL (primary) + TimescaleDB extension for time-series earnings data
Job Queue	Redis + Celery (payout queue, retry logic, batch sweep)
Real-time Events	Supabase Realtime (WebSockets for claim status push)

ML / AI

Component	Technology
Baseline Engine	XGBoost + scikit-learn, served via FastAPI ML endpoint
NLP Classifier	IndicBERT fine-tuned, hosted on HuggingFace Inference API
Fraud Score	LightGBM ensemble, retrained weekly via Airflow DAG
Vision Review	CLIP + custom CNN for obstruction classification
LLM Claim Assessment	Claude API (claude-sonnet-4-20250514)
Feature Store	Feast (offline + online)
Experiment Tracking	MLflow

Data Integrations

Source	Purpose
IMD District API	Rainfall, cyclone, fog triggers
CPCB AQI API	Air quality triggers
Google Maps Platform (Routes + Traffic)	Zone traffic speed corroboration
Windy.com API	Hyperlocal 5-minute rainfall radar

Source	Purpose
News RSS feeds (40+ outlets)	Bandh / protest NLP classification
Twitter/X Filtered Stream	Real-time social disruption signals
Razorpay Payouts / Cashfree	UPI disbursement + NEFT fallback
Digilocker API	Aadhaar-based KYC
NPCI BBPS (future)	Premium auto-debit
Platform Partner APIs (Blinkit, Zepto)	Dark store assignment rate, rider shift data

Infrastructure

Layer	Technology
Hosting	AWS (Mumbai region — ap-south-1)
Containerisation	Docker + ECS Fargate
CI/CD	GitHub Actions → ECR → ECS rolling deploy
Monitoring	Datadog (APM + log aggregation)
Secrets	AWS Secrets Manager
CDN	CloudFront

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7. Development Plan

Phase 1 — MVP (Weeks 1–10)

Goal: One city (Bengaluru), one platform (Blinkit), top 5 auto-triggers only. 200 rider beta.

Milestone	Week
Rider onboarding flow (KYC + bank verification)	W1–2
Earnings baseline engine (rule-based, pre-ML)	W2–3
IMD + CPCB API integrations + auto-trigger classifier	W3–4
Dark store assignment rate integration (Blinkit API)	W4–5
Claim processing engine + UPI payout flow	W5–6
Rider app (React Native) — core screens	W5–7
Fraud firewall — peer consensus + geofence checks	W7–8
Beta onboarding — 200 riders, Bengaluru	W8–9
Monitoring, error handling, payout reconciliation	W9–10

Phase 2 — Expansion (Weeks 11–20)

Goal: 3 cities, all 3 platforms, semi-auto triggers, ML fraud model.

Milestone	Week
NLP disruption classifier (news + Twitter)	W11–13
Semi-auto trigger lane + one-tap rider flow	W12–14
ML-based earnings baseline model	W13–15
LightGBM fraud score (trained on Phase 1 data)	W14–16
Manual claim lane (video upload + AI review)	W15–17
Chennai + Hyderabad city onboarding	W16–18
Zepto + Instamart platform integrations	W17–19
Premium repricing engine (4-week recalibration)	W18–20

Phase 3 — Scale & Compliance (Weeks 21–30)

Goal: IRDAI regulatory compliance, 5 cities, 10,000 riders, reinsurance integration.

Milestone	Week
IRDAI Sandbox application + compliance audit	W21–24
Reinsurance partner API integration	W23–26
Delhi NCR + Mumbai onboarding	W24–27
Full actuarial pricing model deployment	W25–28
Native Android app (Play Store release)	W26–29
Loss ratio reporting dashboard for reinsurance partners	W28–30

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8. Risk, Compliance & Regulatory Notes

IRDAI Classification: SHIELD operates as a parametric insurance product. Under IRDAI's Regulatory Sandbox framework (Insurance Regulatory and Development Authority of India, 2019 Sandbox Regulations), parametric products targeting gig workers are explicitly listed as a priority innovation category. Phase 1 and 2 are operable under the Sandbox exemption; full IRDAI licence is required before Phase 3 scale.

GST Treatment: Payouts are classified as insurance claim settlements (not commercial income transfers). The payment API remarks field explicitly labels each transfer `INSURANCE_CLAIM_SETTLEMENT` to ensure correct tax treatment at the rider's bank and avoid TDS triggers at the 2% threshold.

Data Privacy: Rider GPS data is retained for 90 days for fraud audit purposes only, then anonymised. Location data is never sold or shared with delivery platforms. DPDP Act 2023 compliance is built into the data architecture from day one — explicit consent collected at onboarding for each data category.

Reinsurance: A minimum 40% of premium pool is ceded to a reinsurance partner (Munich Re or Swiss Re India, both active in Indian parametric space) from Phase 3 onwards, protecting the platform against correlated city-wide loss events (a cyclone causing 80% of Chennai riders to claim simultaneously).

Rider Consent & Transparency: Riders receive a plain-language, two-page policy document in their preferred language (English, Kannada, Tamil, Telugu, Hindi) at onboarding. The weekly premium, maximum weekly payout, and top 5 trigger events are displayed on the app home screen — always visible, never buried.

Contributing

This repository contains the core SHIELD product specification, algorithm design, and development roadmap. For technical contribution guidelines, see `CONTRIBUTING.md`.

For partnership inquiries (delivery platform API integration, reinsurance), contact: `partnerships@shieldgig.in`

SHIELD is being built for India's 12 million+ Q-commerce and food delivery riders — the people who absorb the last-mile risk so that a city can have groceries in 10 minutes.